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Microarray Analysis Gene Expression Profiles in Laryngeal Muscle After Recurrent Laryngeal Nerve Injury

Khadijeh Bijangi-Vishehsaraei, PhD[1], Kevin Blum, BS[2], Hongji Zhang, MD[1], Ahmad R. Safa, PhD[1], and Stacey L. Halum, MD[3]

Abstract

Objectives: The pathophysiology of recurrent laryngeal nerve (RLN) transection injury is rare in that it is characteristically followed by a high degree of spontaneous reinnervation, with reinnervation of the laryngeal adductor complex (AC) preceding that of the abducting posterior cricoarytenoid (PCA) muscle. Here, we aim to elucidate the differentially expressed myogenic factors following RLN injury that may be at least partially responsible for the spontaneous reinnervation.

Methods: F344 male rats underwent RLN injury (n = 12) or sham surgery (n = 12). One week after RLN injury, larynges were harvested following euthanasia. The mRNA was extracted from PCA and AC muscles bilaterally, and microarray analysis was performed using a full rat genome array.

Results: Microarray analysis of denervated AC and PCA muscles demonstrated dramatic differences in gene expression profiles, with 205 individual probes that were differentially expressed between the denervated AC and PCA muscles and only 14 genes with similar expression patterns.

Conclusions: The differential expression patterns of the AC and PCA suggest different mechanisms of reinnervation. The PCA showed the gene patterns of Wallerian degeneration, while the AC expressed the gene patterns of reinnervation by adjacent axonal sprouting. This finding may reveal important therapeutic targets applicable to RLN and other peripheral nerve injuries.

Keywords

vocal fold, recurrent laryngeal nerve, laryngeal muscles, larynx, denervation

Introduction

Recent studies based on animal models have suggested that a high rate of spontaneous reinnervation ensues after recurrent laryngeal nerve (RLN) injury,[1-7] with the adductor muscle complex (AC) receiving preferential reinnervation that precedes reinnervation of the single abductor, the posterior cricoarytenoid (PCA) muscle.[1,2] In such animal models, studies have suggested that the reinnervation after RLN injury is due to intramuscular axonal sprouting via the superior laryngeal nerve (SLN), a process that may be temporally and quantitatively different between the AC and PCA.[2-4,6] No previous studies have investigated the differences in gene expression profiles between the AC and PCA after RLN injury in the rodent model, yet elucidating the local myogenic changes is critical to better understanding the incongruent reinnervation patterns. Such findings may also help elucidate future therapeutic targets for the clinical management of RLN injury.

In the current investigations, we hypothesized that the spontaneous reinnervation of laryngeal muscles that occurs after RLN injury is associated with alterations in gene expression from the denervated laryngeal muscles that can be qualitatively and quantitatively determined using microarray analysis. Furthermore, based on different patterns in spontaneous reinnervation of the AC and PCA, we hypothesized that differences would be detected in gene expression profiles of the denervated PCA profiles when compared to the denervated laryngeal AC.

Because of the extensive number of factors expected to be differentially expressed after RLN injury, microarray

[1]Department of Pharmacology and Toxicology, Indiana University, Indianapolis, Indiana, USA
[2]Purdue University Weldon School of Biomedical Engineering, West Lafayette, Indiana, USA
[3]Department of Speech, Language, and Hearing Sciences, Purdue University, West Lafayette, Indiana, USA

Corresponding Author:
Stacey L. Halum, MD, Department of Speech, Language, and Hearing Sciences, Purdue University, West Lafayette, IN 47907, USA.
Email: shalum@voiceindy.com

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analysis was selected as an initial screening tool for this experiment to permit qualitative and quantitative analysis of thousands of gene products simultaneously rather than restricting the investigation to known gene products of interest. Application of the microarray approach permits detection of more elusive factors or those expressed at low levels. Furthermore, the comparative approach facilitates detection of factors that are differentially expressed in the AC and PCA after RLN injury, thus potentially revealing important gene expression differences between the AC and PCA that could help explain their incongruent reinnervation patterns.[1-6] While microarray could be customized for neurotrophic factor (NF) and neurotrophic factor receptor (NFR) genes only, this would prevent identification of many other factors that play a role in the changes that follow RLN injury. For example, cytokines and inflammatory mediators such as the interleukins and tumor necrosis factor superfamily may have indirect influence on RLN regeneration via immune response modulation as peripheral nerve regeneration has been shown to be directly impacted by macrophage activity, which is dependent on such immune modulators. As another example, apoptotic and anti-apoptotic factors likely play a significant role after axotomy as peripheral nerve injury is characteristically followed by severe apoptosis of motoneurons; furthermore, loss of apoptic factors such as p53, bax, and Fas has been shown to protect from motoneuron apoptosis after axotomy.[8-11] Thus, the microarray approach permits identification of a wide range of gene expression changes beyond the changes in what would traditionally be considered NFs, with the data derived being hypothesis generating in nature.

Methods

Animal Model of RLN Injury

A total of 24 F344 rats were used. All animals underwent anesthesia with intraperitoneal (IP) ketamine (75 mg/kg) and xylazine (10 mg/kg). Study animals underwent unilateral RLN transection (n = 12) with removal of a 1 cm nerve segment, while control animals underwent the same surgical approach with no nerve injury (n = 12). Before awakening from anesthesia, unilateral vocal fold motion immobility and normal motion were confirmed in the study and control animals, respectively. At 1 week following the denervation surgery, all 12 animals from each group were euthanized. This time point was selected because internal and microenvironmental changes involving denervated laryngeal muscle have been demonstrated within 1 week after denervation injury.[12-14] The larynges and distal 1 cm of the transected RLN were harvested from each animal, with dissection of the denervated or control hemi-laryngeal muscles from the thyroid cartilage and separation of the AC muscles from the laryngeal abductor, the PCA. The AC included the thyroarytenoid (TA) muscle (lateral and medial TA), the alar cricoarytenoid muscle (ACA), and the lateral cricoarytenoid muscle (LCA).[15] The AC and PCA were evaluated separately. Harvested muscle and nerve were immediately placed in labeled tubes with RNA stabilizing agent (RNAlater, Sigma-Aldrich, Austin, Texas, USA); specimens were stored at −80°C until the time of RNA purification.

Tissue Homogenization and RNA Purification

For each adductor or PCA specimen, 30 to 50 mg of tissue was disrupted and homogenized using zirconia 1 mm beads (BioSpec Products Inc, Bartlesville, Oklahoma, USA) with density of 3.7 g/ml in a tissue homogenizer (Min-bead Beater-8, BioSpec Products Inc) for 3 minutes at 4°C. The total RNA was extracted and purified using RNeasy Fibrous Tissue MiniKit (QIAGEN, Austin, Texas, USA). The RNA concentrations were determined using a NanoDrop ND-1000 Spectrophotometer (Thermo Fisher Scientific Inc, Waltham, Massachusetts, USA). The quality of purified total RNA was determined by optical density (A_{260}/A_{280} ratio between 1.8 and 2.1) and electrophoresis on 1% agarose gel (ensuring the 18S and 28S bands are clearly distinguished).

Microarray Processing and Analysis

The microarray experiments were carried out using the facilities of the Center for Medical Genomics at Indiana University School of Medicine, which was funded in part by a grant from the Indiana 21st Century Research and Technology Fund and by the Indiana Genomics Initiative (INGEN). (INGEN is supported in part by the Lilly Endowment.) Total RNA samples were labeled using the standard protocol for the Ambion WT Expression kit combined with the Affymetrix GeneChip WT Terminal Labeling and Controls Kit using 100 ng of total RNA. Individual labeled samples were hybridized to the Rat Gene 1.0 ST GeneChips for 17 hours then washed, stained, and scanned with the standard protocol using Affymetrix GeneChip Command Console Software (AGCC) to generate data (CEL files). Arrays were visually scanned for abnormalities or defects. The CEL files were imported into Partek Genomics Suite (Partek, Inc, St Louis, Missouri, USA). Robust multi-array average (RMA) signals were generated for the core probe sets using the RMA background correction, Quantile normalization, and summarization by Median Polish.[16,17] Summarized signals for each probe set were log2 transformed. These log transformed signals were used for principal components analysis, hierarchical clustering, and signal histograms to determine if there were any outlier arrays (no outliers were detected). Untransformed RMA signals were used for fold change calculations. Data were analyzed using a 2-way ANOVA with factors for tissue (PCA, AC) and treatment (sham, RLN) using the log2

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Table 1. Gene Products Upregulated From Both the AC and PCA Muscles After RLN Injury When Compared to AC and PCA Sham Controls, Respectively.

General Category	Upregulated Factors	PCA Fold	AC Fold	Summarized Role
	Glycoprotein (transmembrane)	14.62	3.31	Transmembrane proteins
Muscle associated	Transmembrane protein 8C	6.62	11.78	
	ADAM metallopeptidase domain 12	6.58	4.54	
	Cholinergic receptor, nicotinic α1	3.89	7.2	Neuromuscular junction receptor
	Myosin HC3	3.73	5.95	Contractile function
	Troponin T2	3.32	3.99	
Vascular associated	Collagen triple helix repeat 1	6.1	3.11	Angiogenesis
	Matrix metallopeptidase 12	22.66	5.11	
	Leukocyte Ig-like receptor Gp49B	13.07	5.66	
Inflammatory	Cystatin F (leukocystatin)	4.19	3.58	Immune regulation and nonspecific inflammatory mediators
	Kallikrein 1-like peptidase	4.08	3.61	
	Dendritic cell inhibitory receptor	3.56	3.07	
	Downregulated Factors			
Muscle associated	Myostatin (Mstn)	−5.74	−3.36	Skeletal muscle atrophy
Other	Family with sequence similarity 55b (Fam 55b)	−3.87	−3.13	Unknown

Abbreviations: AC, adductor complex; PCA, posterior cricoarytenoid; RLN, recurrent laryngeal nerve.

transformed expression levels. Contrasts were done for PCA treated versus PCA control and AC treated versus AC controls. Fold changes were calculated using the untransformed RMA signals. Probe sets with log2 expression levels <4.0 were close to background. Probe sets with average expression levels <4.0 for all 4 groups were removed before the false discovery rate (FDR) was calculated using the Storey qvalue method.[18] Gene products were categorized according to function based on Gene Ontology and DAVID[19] (see Table 1).

Real-Time Polymerase Chain Reaction. To validate the differential expression of representative genes from each functional category of gene expression (see Table 1), quantitative real-time polymerase chain reaction (RT-PCR) was performed using an ABI PRISM 7700 Sequence Detection System (Applied Biosystems, Foster City, California, USA). The iScript 1-step qRT-PCR kit with SYBR Green (Bio-Rad, Life Science Research, Hercules, California, USA) was used for cDNA synthesis and PCR amplification according to manufacturer's instruction using target gene-specific primer pairs (see Table 5 for gene specific primers). The amount of target gene transcript normalized to the endogenous glyceraldehyde 3-phosphate dehydrogenase (GAPDH) housekeeping gene transcript was computed based on a comparative cycle threshold method as instructed by Applied Biosystems 7500/7500 Fast manuals. Briefly, 50 ng of RNA in total volume of 25 µl was reverse transcripted (cDNA synthesis) followed by reverse transcription inactivation for 5 minutes at 95°C. In presence of target gene primers (at 300 nM final concentration/sense and antisense primers), the amplification conditions consisted of 40 cycles of denaturation at 95°C for 10 seconds, annealing at 60°C for 30 seconds, and extension at 72°C for 10 seconds, followed by melting curve analysis. SYBR green is a DNA-binding dye that incorporate into dsDNA.

Results

Gene Regulation Summary

In total, there were 36 gene products in the AC and 219 gene products in the PCA that were up- or downregulated after RLN injury more than 3-fold relative to sham controls. Genes with known roles were classified based on their generalized function, while those without defined function were classified as "unknown." A summary of the roles of the gene products regulated in the RLN injury AC relative to AC sham controls can be found in Figure 1. A summary of the roles of the gene products regulated in the RLN injury PCA relative to PCA sham controls can be found in Figure 2. For a summary of the fold changes in the gene roles seen in each muscle group, see Figure 3.

Genes Similarly Regulated in AC and PCA

A total of 14 gene products were similarly regulated in both the RLN injured AC and PCA relative to respective sham

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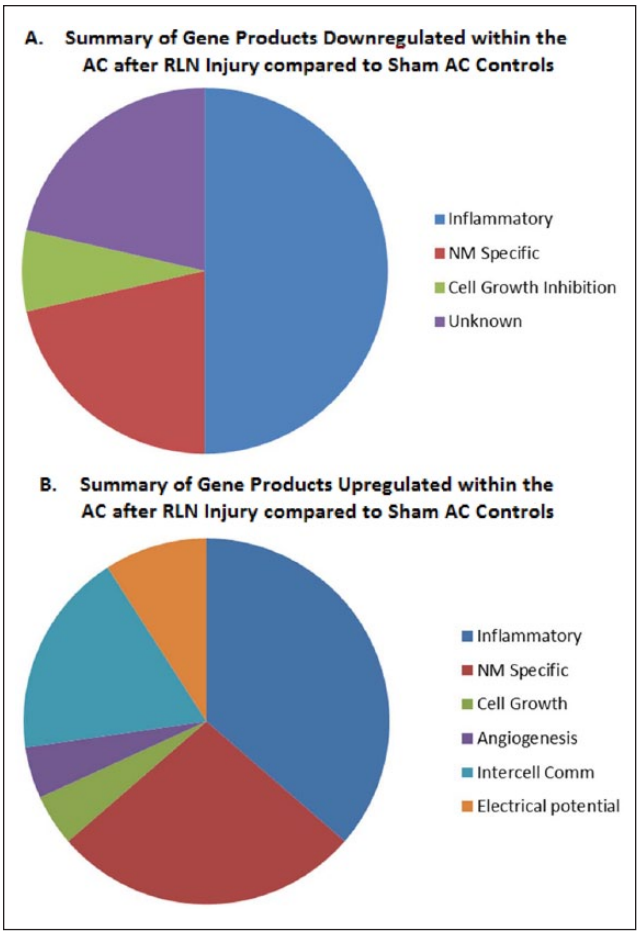


Figure 1. Changes in adductor complex (AC) gene expression after recurrent laryngeal nerve (RLN) injury versus sham AC controls. A total of 36 gene products in the AC muscle were differentially expressed after RLN injury when compared with the AC controls, with 22 products being upregulated and 14 products being downregulated. (A) Upregulated gene products were categorized according to roles, with upregulated products being inflammatory (8), neuromuscular (NM) specific (6), cell growth related (1), associated with angiogenesis (1), mediating intercellular communication (4), or maintenance of electrical potential (2). (B) Downregulated gene products were categorized according to roles, with downregulated products being inflammatory (7), NM specific (3), cell growth inhibition related (1), or having unknown roles (3).

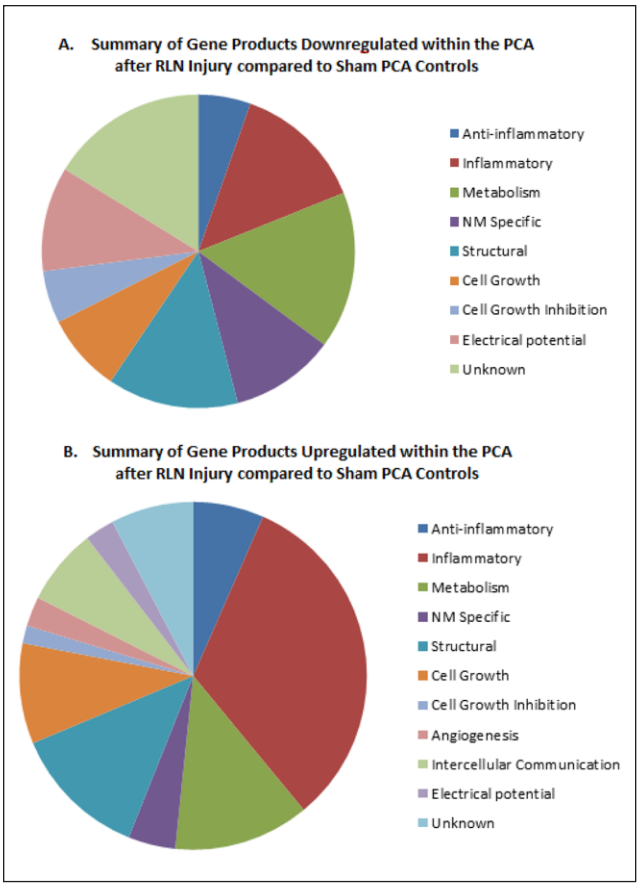


Figure 2. Changes in posterior cricoarytenoid (PCA) gene expression after recurrent laryngeal nerve (RLN) injury versus sham PCA controls. A total of 219 gene products in the PCA muscle were differentially expressed after RLN injury when compared with the PCA controls, with 182 products being upregulated and 37 products being downregulated. (A) Upregulated gene products were categorized according to roles, with upregulated products being inflammatory (59), anti-inflammatory (12), metabolic (23), neuromuscular (NM) specific (8), structural (23), cell growth related (17), cell growth inhibition related (3), associated with angiogenesis (5), mediating intercellular communication (13), maintenance of electrical potential (5), or having unknown roles (14). (B) Downregulated gene products were categorized according to roles, with downregulated products being inflammatory (5), anti-inflammatory (2), metabolic (6), NM specific (4), structural (5), cell growth related (3), cell growth inhibition related (2), maintenance of electrical potential (4), or having unkown roles (6).

controls, including 12 upregulated gene products and 2 downregulated gene products. The upregulated genes were comprised of 6 muscle-, 1 vascular-, and 5 inflammatory-related gene products. The downregulated genes were 1 muscle-related gene product and 1 gene product with an unknown role. While the direction of regulation was consistent between the AC and the PCA, the extent of regulation was similar in some cases but drastically different in others. For example, Kallikrein 1–like peptidase showed a fold change of 4.08 in the PCA and 3.61 in the AC, while matrix metalopeptidase 12 showed a fold change of 22.66 in the PCA and 5.11 in the AC. Details about the specific genes products and their roles can be found in Table 1.

Genes Regulated in AC Only

There were 10 gene products that were upregulated in the RLN injury AC relative to sham controls that were not

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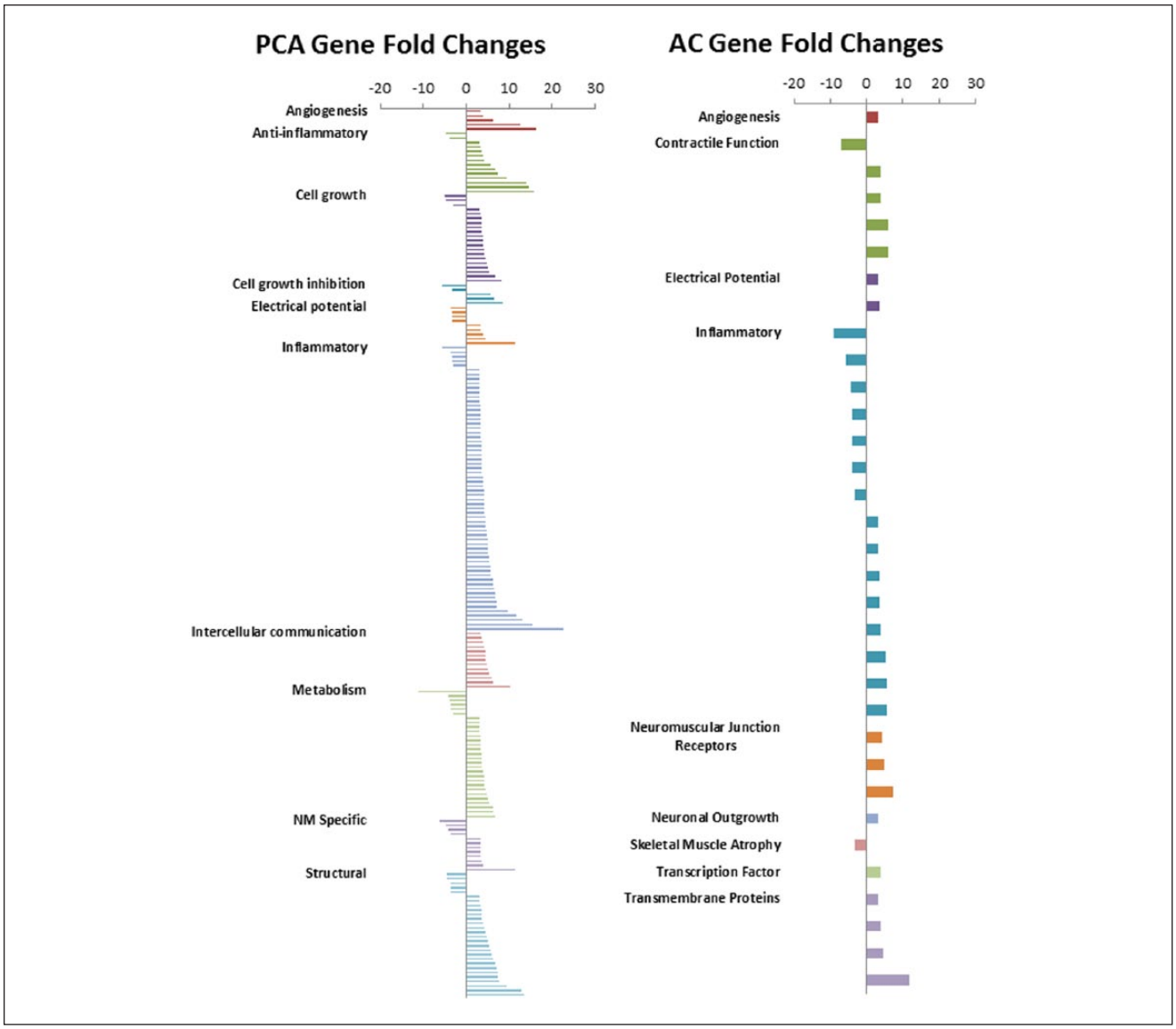


Figure 3. Summarized gene expression fold changes 1 week after recurrent laryngeal nerve (RLN) injury versus sham posterior cricoarytenoid (PCA) controls. A total of (A) 219 gene products in the PCA muscle and (B) 36 gene products in the adductor complex (AC) muscle were differentially expressed after RLN injury compared to sham controls. Gene products were categorized according to roles, and the fold change in each gene product is presented. Those gene products with unknown roles are not shown here.

upregulated in the RLN injury PCA relative to sham controls. The genes included 6 muscle-, 3 neuron-, and 1 inflammatory-related gene product. None of the fold changes were greater than 6. Details about the specific genes products and their roles can be found in Table 2.

There were 10 genes that were downregulated in the RLN injury AC that were not downregulated in the RLN injury PCA relative to sham controls. The gene products affected were 1 muscle- and 7 inflammatory-related gene products and 2 gene products with unknown roles. The downregulation fold change ranged from −3.32 to −9.14. Details about the specific genes and their roles can be found in Table 2.

Genes Regulated in PCA Only

There were 182 gene products that were upregulated in only the RLN injury PCA relative to PCA sham controls 1 week after RLN injury. The largest subsection was inflammatory-related gene products, with 54 gene products upregulated in the RLN injury PCA relative to sham PCA controls. Several anti-inflammatory gene products were

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Table 2. Differentially Expressed Gene Products From the AC After RLN Injury Relative to AC Sham Controls.[a]

General Category	Upregulated Factors	AC Fold[b]	Summarized Role
	Cholinergic receptor, nicotinic delta	4.97	Neuromuscular junction receptors
	Cholinergic receptor, nicotinic gamma	4.17	
Muscle associated	Myogenin	3.92	Transcription factor
	Organic solute transporter α	3.73	Transmembrane proteins
	Calsequestrin 2	3.59	Electrical potential maintenance
	Sodium channel voltage gated II α subunit	3.25	
Neuron associated	Neural cell adhesion molecule I (NCAM-1)	3.07	Neuronal outgrowth
	Leukocyte Ig-like receptor Gp49B	5.66	Immune regulation and nonspecific inflammatory mediators
	FcReceptor-like S	3.95	
Inflammatory	Interleukin 1 beta	3.3	
	Downregulated Factors		
Muscle associated	Myosin heavy chain 13, skeletal muscle	−7.06	Contractile function
	Neutrophilic granule protein	−9.14	Immune regulation and nonspecific inflammatory mediators
	Myosin binding protein H-like	−5.6	
	Myeloperoxidase	−4.36	
Inflammatory	Defensin rat NP-3 and NP-4 precursor	−4.07	
	Defensin, alpha 5, paneth cell-specific	−3.98	
	Similar to glycophorin	−3.86	
	Cytochrome p450 family, A,3a	−3.32	
	Eraf erythroid associated factor	−6.23	Unknown
Other	SEC14-like 3	−3.48	

Abbreviations: AC, adductor complex; RLN, recurrent laryngeal nerve.
[a]Each listed gene product demonstrated a statistically significant difference in gene expression when compared to control (sham surgery) gene products ($P < .01$). Factors listed in the table did not show differential expression from the PCA after RLN injury.
[b]A 3-fold over more variation in gene expression was the predetermined threshold for differential expression.

also upregulated. Details about the specific genes that were upregulated or downregulated can be found in Table 3 and Table 4, respectively.

RT-PCR Confirmation

The RT-PCR consistently confirmed both the directional patterns and statistical significance of the differentially expressed gene products identified via microarray analysis. Summary of primers used and graphic results of the confirmatory studies are included in Table 5 and Figure 4, respectively.

Discussion

There is a clinical difference in how the adductor and abductor muscles of the larynx are affected following RLN injury. In the animal model, the AC and PCA demonstrate significantly different innervation patterns and responses to RLN injury.[1-7] Studies have suggested intramuscular axonal sprouting via the SLN may be a source of rapid laryngeal reinnervation after RLN injury. While it remains unclear if such a model holds direct clinical relevance, the authors believe it may help explain the clinical phenomenon of post-RLN injury synkinetic reinnervation. This study sought to examine the underlying genomic factors expressed from laryngeal muscles after RLN injury via microarray analysis. At 1 week post–RLN injury, the AC and PCA muscle groups showed markedly different genomic expression relative to sham controls as well as relative to each other, suggesting a differential response to nerve injury in the abductor and adductor muscles.

To understand these changes in the context of RLN regeneration, the physiologic changes that ensue after RLN injury must first be considered. Immediately following axotomy, Wallerian degeneration of the distal axon begins, with influx of calcium ions and activation of calcium-dependent proteases leading to axonal edema and cytoskeleton breakdown.[10] Within hours of axotomy, surrounding Schwann cells initiate an inflammatory cascade by releasing a wide range of inflammatory cytokines, thereby recruiting macrophages to remove the myelin and axonal debris.[11] Denervated Schwann cells downregulate their expression of structural proteins, resulting in the destruction of their myelin sheaths and further contributing to degenerative debris. During this, macrophages have an essential role as

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Table 3. Selected Upregulated PCA Products After RLN Injury That Are Not Differentially Expressed From the AC After RLN Injury.

Gene Symbol	Name	PCA Fold[a]	Summarized Role
Fcgr1a	Fc fragment of IgG, high affinity Ia, receptor (CD64)	3.16	Inflammatory
Lgmn	Legumain	3.16	Inflammatory
Ccr2, Agtr1b	Chemokine (C-C motif) receptor 2	3.17	Inflammatory
Ly49si2	Immunoreceptor Ly49si2	3.24	Inflammatory
Igsf6	Immunoglobulin superfamily member 6	3.34	Inflammatory
Tm7sf4	Transmembrane 7 superfamily member 4	3.34	Inflammatory
Csf3r	Colony stimulating factor 3 receptor	3.51	Inflammatory
Il8rb	Interleukin 8 receptor, beta	3.6	Inflammatory
Mpeg1	Macrophage expressed 1	3.83	Inflammatory
Prss35	Protease, serine, 35	3.99	Inflammatory
Anxa8	Annexin A8	4.05	Angiogenesis
Lyz2	Lysozyme 2; lysozyme C type 2	4.09	Inflammatory
Ccl3	Chemokine (C-C motif) ligand	4.18	Inflammatory
Il1f9	Interleukin 1 family, member 9	4.22	Inflammatory
Msr1	Macrophage scavenger receptor 1	4.25	Inflammatory
Sgpp2	Sphingosine-1-phosphate phosphatase2	5.06	Inflammatory
Slamf9	SLAM family member 9	5.08	Inflammatory
Tmprss11f	Transmembrane protease, serine 11F	5.37	Inflammatory
Trem2	Triggering receptor expressed on myeloid cells 2	5.49	Inflammatory
Clec4e	C-type lectin domain family 4, member E	5.7	Inflammatory
Ilif5	Interleukin 1 family, member 5 (delta)	5.78	Inflammatory
Cthrc1	Collagen triple helix repeat containing 1	6.1	Angiogenesis
Itgam	Integrin, alpha M (complement component 3 receptor 3)	6.14	Inflammatory
Ccl2	Chemokine (C-C motif) ligand 2	6.17	Inflammatory
Elmod1	ELMO/CED-12 domain containing 1	6.43	Inflammatory
Ccl9	C-C motif chemokine 9	6.68	Inflammatory
Ly6g6c	Lymphocyte antigen 6 complex, locus G6C	7.05	Inflammatory
Klk9	Kallikrein-related peptidase 9	7.27	Inflammatory
Klk10	Kallikrein-related peptidase 10	9.78	Inflammatory
Lrrc15	Leucine rich repeat containing 15	11.58	Inflammatory
Il1rn	Interleukin 1 receptor antagonist	13.95	Anti-inflammatory
Mmp9	Matrix metallopeptidase 9	15.49	Inflammatory
Serpinb12	Serpin peptidase inhibitor, clade B, member 12	16.39	Angiogenesis

Abbreviations: AC, adductor complex; PCA, posterior cricoarytenoid; RLN, recurrent laryngeal nerve.
[a]The gene products were selected from a list of 182 products that were upregulated in the PCA 1 week after recurrent laryngeal nerve (RLN) injury and not altered in the adductor muscle complex (AC) 1 week after RLN injury (*See Figure 2A for role summary*). As demonstrated in this table, the majority of these gene products had an inflammatory role.

they remove debris and protein such as myelin-associated glycoprotein that, if present, would inhibit motoneuron outgrowth. After debris removal, Schwann cells proliferate and align to form columns of cells (bands of Bungner), using contact guidance and key regulatory factors to induce axonal elongation and direct regenerating axons to the endplate target. In addition to directing regenerating injured axons, Schwann cells can form bridges from denervated endplates to innervated endplates, thereby providing a mechanism for non-injured axons to participate in motoneuron sprouting and reinnervation of adjacent denervated endplates.[8,11,12] Neurotrophic factors (NFs) are upregulated at the muscle end target and along the Schwann cell bridges to promote and guide axonal outgrowth. The NFs refer to not only the neurotrophins NGF, BDNF, NT-3, and NT-4/5 that act on the trk family of tyrosine kinase receptors, but they also include other neuropeptides that provide neuronal support. Based on this definition, the list of potential NFs is exhaustive and includes proteins such as the GDNF family of ligands (GDNF, neurturin [NTN], artemin [ART], and persephin [PSP]), neuregulin 1 with its isoforms, melanocortins, and a number of other growth factors such as ciliary neurotrophic factor (CNTF), insulin-like growth factor-1 (IGF-1), transforming growth factors (TGFa and TGFb), VGF peptide, glia maturation factors (GMFs), and fibroblast growth factors (FGF 1, 2, 9).[9,20] Interestingly, our study did not reveal

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Table 4. Selected Downregulated PCA Products After RLN Injury That Are Not Differentially Expressed From the AC After RLN Injury.

Gene Symbol	Name	PCA Fold[a]	Summarized Role
Ucp1	Uncoupling protein 1 (mitochondrial, proton carrier)	−10.99	Metabolism
Myh7	Myosin, heavy chain 7, cardiac muscle, beta	−6.06	Muscle specific
Mybphl	Myosin binding protein H-like	−5.48	Inflammatory
Scrg1	Stimulator of chondrogenesis 1	−5.03	Cell growth
Fabp3	Fatty acid binding protein 3, muscle and heart (mammary-derived growth	−4.87	Muscle specific
Serpina6	Serpin peptidase inhibitor, clade A	−4.72	Anti-inflammatory
Glt25d2	Glycosyltransferase 25 domain containing 2	−4.71	Cell growth
Clec3a	C-type lectin domain family 3, member A	−4.46	Structural
Col10a1	Collagen, type X, alpha 1	−4.35	Structural
Pck1	Phosphoenolpyruvate carboxykinase 1	−4.12	Metabolism
Myl2	Myosin, light chain 2, regulatory, cardiac, slow	−4.12	Muscle Specific
Stk32b	Serine/threonine-protein kinase 32B	−3.9	Metabolism
A1i3, Mug2, RGD1566313	Alpha-1-inhibitor III	−3.76	Anti-inflammatory
Acan	Aggrecan	−3.74	Structural
Hlapln1	Hyaluronan and proteoglycan link protein 1	−3.67	Structural
Pfkfb1	6-phosphofructo-2-kinase/fructose-2,6-biphosphatase	−3.65	Metabolism
Matn1	Matrilin 1, cartilage matrix protein	−3.65	Structural
myh2, MYH1	Myosin, heavy chain 1, skeletal muscle, adult	−3.63	NM specific
Pld5	Phospholipase D family, member 5.	−3.54	Metabolism
Scn4b	Sodium channel, voltage-gated, type IV, beta	−3.49	Electrical potential
Asb14	Ankyrin repeat and SOCS box-containing 14	−3.49	Inflammatory
Ucma	Unique cartilage matrix-associated protein	−3.43	Cell growth inhibition
Aqp7	Aquaporin 7	−3.37	Electrical potential
Kcnma1	Potassium large conductance calcium-activated channel, subfamily M, alpha member 1	−3.32	Electrical potential
Padi2	Peptidyl arginine deiminase, type II	−3.28	Inflammatory
Lect1	Leukocyte cell derived chemotaxin 1	−3.24	Inflammatory
Clcn1	Chloride channel 1, skeletal muscle	−3.18	Electrical potential
Smgc	Submandibular gland protein C	−3.11	Cell growth
Gpd1	Glycerol-3-phosphate dehydrogenase 1 (soluble)	−3.04	Metabolism
Bpil3	Bactericidal/permeability-increasing protein-like 3	−3.01	Inflammatory

Abbreviations: AC, adductor complex; PCA, posterior cricoarytenoid; NM, neuromuscular; RLN, recurrent laryngeal nerve.
[a]The gene products were selected from a list of 37 products that were downregulated in the PCA after RLN injury and not altered in the adductor muscle complex after RLN injury. (See Figure 2A for role summary.)

differential expression of any of these well-characterized NFs, which likely has to do with the timing of the analysis.

Despite the lack of differential NF expression in this study, the information gleaned helps elucidate the specific mechanisms at work in the AC and PCA muscles after RLN injury. First, the microarray profiles of both the AC muscles and PCA muscles did support that an acute denervation injury had occurred, in that the RLN injury induced gene expression changes characteristic of acute muscle denervation. For example, there was upregulation of myogenin and downregulation of myostatin, which are both signs of acute denervation.[21-24] Furthermore, cholinergic receptor nicotinic alpha 1, which classically surges after acute denervation,[25] dramatically upregulated in both the AC and PCA, supporting that the RLN transection procedure had indeed disrupted a primary innervation source.

Despite these similarities, the overall response to acute loss of RLN innervation was extremely different between the AC and PCA. Most notably, after RLN injury, the PCA (relative to sham PCA) demonstrated 205 individual probes that had a 3-fold or greater change and were not differentially expressed from the denervated AC (relative to sham AC). The majority of these factors were upregulated inflammatory mediators of muscle atrophy and Wallerian degeneration. For example, the

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Table 5. Primer Pairs Used for Real-Time Polymerase Chain Reaction.

Gene	Sense	Anti-sense
Gpnmb	5'-CCCCAGGCTCCTTAGATTTC-3'	5'-GTTGATGAAGCATGGCCTTT-3'
Chrna1	5'-TTGTCACCCACTTTCCCTTC-3'	5'-TCTCCGCTCTCCATGAAGTT-3'
Lilrb4	5'-TCCTTGGGACACTCAACTCC-3'	5'-GGCATCACTGTGCTCTGAAA-3'
Mstn	5'-ACGCTACCACGGAAACAATC-3'	5'-AGCTGGGCCTTTACCACTTT-3'
Mmp12	5'-TGCAGCTGTCTTTGATCCAC-3'	5'-GCATCAATTTTTGGCCTGAT-3'
Il1b	5'-CAGGAAGGCAGTGTCACTCA-3'	5'-AAAGAAGGTGCTTGGGTCCT-3'
Mmp9	5'-GCCGGATACAAACTGGTGAT-3'	5'-TGGGGGATCCGTGTTTATTA-3'
GAPDH	5'-AGACAGCCGCATCTTCTTGT-3'	5'-TGATGGCAACAATGTCCACT-3'

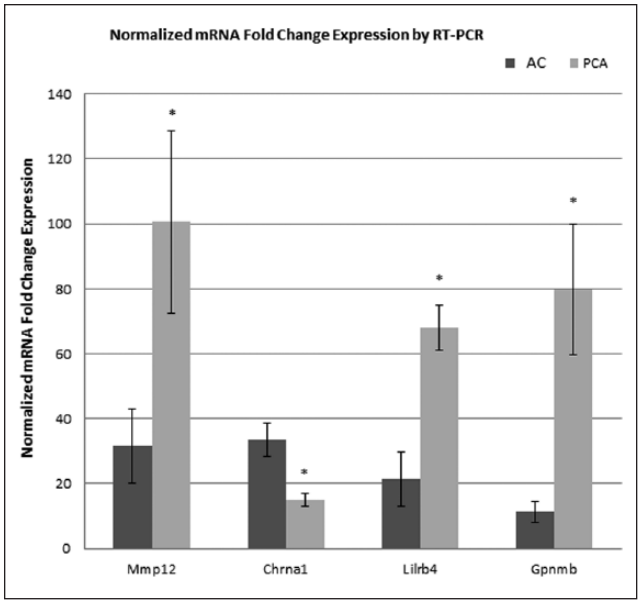


Figure 4. Normalized mRNA fold change expression by real-time polymerase chain (RT-PCR) reaction in adductor complex (AC) and posterior cricoarytenoid (PCA). Confirmatory RT-PCR was performed on multiple gene products that demonstrated differential expression by microarray analysis. The absolute mRNA values are expressed here relative to one another. Note that each of these products demonstrated significantly different expression levels between the AC and PCA.
*$P < .05$.

PCA muscle showed extensive upregulation of several protease-related genes, including PRSS35, TMPRSS11D, TMPRSS11F, and TMPRSS13, all of which are serine proteases. These gene products lead to the degradation of the matrix, which has been shown to be a consistent feature in denervation-induced muscular atrophy.[26] The extensive upregulation of cytokines and macrophage-associated proteins (see Table 3) seen in the PCA also suggests that Wallerian degeneration process may be active within the PCA following RLN injury. The fact that the AC demonstrated no such changes suggests it is not undergoing extensive intrinsic changes, perhaps due to an immediate intramuscular axonal sprouting process that "bridges" the AC from an acute denervation status to a rapid reinnervation phase. In fact, studies of rats that have had nerves severed and immediately re-anastomosed have shown, among other genes, upregulation of calsequestrin 2 and troponin, both of which were upregulated in the AC in this study.[24] Such a model would support literature in a rodent model that has suggested that the larynx receives dual innervation from the RLN and SLN,[3-6] with SLN branches mediating rapid reinnervation of the laryngeal myofibers after RLN injury via intramuscular axonal sprouting. However, 1 previous study has suggested that the PCA receives reinnervation via the SLN via intramuscular sprouting,[4] so perhaps our findings in the AC are merely a reflection of the early post-RLN injury time point (1 week) used in this study. Due to potential differences between rat and human physiology, it is important to note that the differences seen may be due to neuroanatomical differences between rats and humans. The relation between rat and human physiology in this context remains to be elucidated and is crucial to the clinical application of the information in this study. However, the authors feel that SLN-derived intramuscular sprouting could explain many post-RLN injury clinical manifestations such as post-injury spontaneous medialization of the immobile vocal fold, persistent false vocal fold adduction, and electromyography-documented synkinesis. Future patient-based studies would be beneficial in establishing the gene expression profile in human laryngeal muscles after RLN injuries with differing severity, reinnervation status (synkinesis, absent reinnervation, etc), and timing.

Declaration of Conflicting Interests

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